جامعة الأخوين

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AL AKHAWAYN UNIVERSITY

School of Science and Engineering

Milestone 3: Data Acquisition Validation and Preparation

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~ Spring 2026 ~

Table of Contents

1. Introduction 3
2. Project Context & Integration with Previous Milestones 3
3. Raw Data Acquisition & Repository Storage 4
   3.1 Dataset Description 4
   3.2 Repository Organization 4
   3.3 Ingestion Pipeline Design 5
   3.4 Output Artifacts 5
4. Data Preprocessing & Transformation 6
   4.1 Objective 6
   4.2 Parsing Challenges and Solutions 6
   4.3 Preprocessing Pipeline 6
   4.4 Output Tables and scale 6
5. Feature Engineering Design 7
   5.1 Objective 7
   5.2 Feature Engineering Pipeline 7
   5.3 Feature Categories 8
   5.4 Feature Summary Table 8
6. Schema Definition 9
   6.1 Objective 9
   6.2 Schema Layers 9
   6.3 Schema Structure 9
   6.4 What the Schema Enforces (Implementation-Level) 10
7. Data Validation & Anomaly Analysis 10
   7.1 Objective 10
   7.2 Validation Pipeline 11
   7.3 Validation Logic 11
   7.4 Outputs 11
8. Data Versioning with DVC 12
   8.1 Objective 12
   8.2 DVC Setup 12
   8.3 Tracked Data Artifacts 12
   8.4 DVC Execution and Flow: 12
9. Feature Store Setup: Feast 13
   9.1 Objective 13
   9.2 Feature Store Components 13
   9.3 Retrieval Example 14
10. Results & Demonstration: Pipeline Execution Results 14
12. Limitations & Next Steps 16
   12.1 Limitations 16
   12.2 Future Work 16

AI-Assisted Symbolic Optimization for Strategic Facility Network Design

1. Introduction

The Uncapacitated Facility Location Problem (UFLP) is a fundamental combinatorial optimization problem in which the objective is to determine which facilities to open and how to assign customers in order to minimize the total cost composed of fixed facility opening costs and customer assignment costs. While classical optimization solvers can efficiently solve UFLP once a mathematical model is defined, a major bottleneck in practice lies in the formulation, preparation, and management of the input data required to build these models.

This project introduces an AI-assisted symbolic optimization pipeline in which a Large Language Model (LLM) generates formal optimization models, while a classical solver ensures correctness and optimality. In this context, Milestone 3 focuses on building a robust data and MLOps layer, enabling structured data ingestion, preprocessing, validation, versioning, and feature management.

Building upon the problem formulation and system architecture defined in Milestone 1 and the solver pipeline implemented in Milestone 2, this milestone ensures that all downstream optimization components operate on validated, reproducible, and reusable data assets.

2. Project Context & Integration with Previous Milestones

The overall system evolves across three complementary layers:

- **Milestone 1 (Problem Framing):**

Defined UFLP as an ML-assisted optimization problem, highlighting the modeling bottleneck and introducing the LLM-as-modeler paradigm

- **Milestone 2 (Solver Layer):**

Implemented:

  - a deterministic OR-Tools MILP baseline
  - an LLM-generated solver pipeline with verification, ensuring correctness through solver execution rather than probabilistic reasoning

- **Milestone 3 (Data & MLOps Layer)**

Introduces:

  - structured data pipelines
  - validation and schema enforcement
  - feature engineering
  - versioning and reproducibility
  - feature store integration

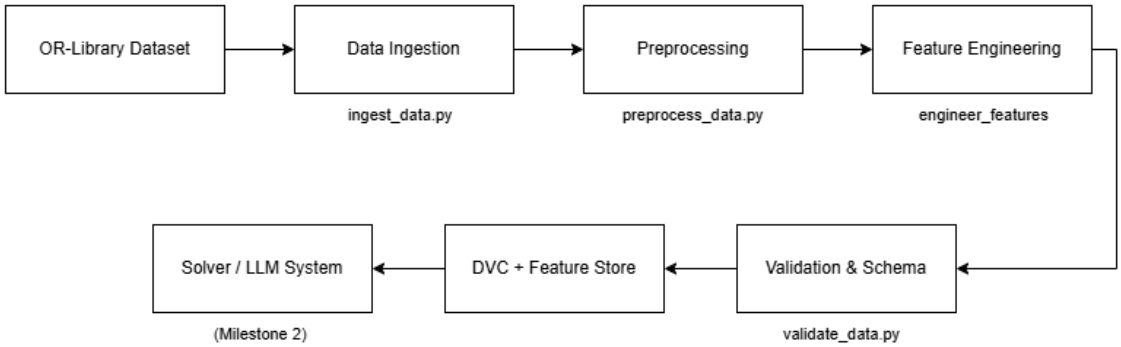


Figure 1: End-to-end architecture integrating data pipeline and solver system

Milestone 3 converts the system from a solver-centric prototype into a data-centric MLOps system, where:

- all inputs are structured and validated
- all transformations are reproducible
- features are reusable across experiments

3. Raw Data Acquisition & Repository Storage

3.1 Dataset Description

The project uses the OR-Library UFLP benchmark dataset, a standardized collection of 15 instances widely used in operations research. Each instance is stored as a text file and contains:

- number of facilities (m)
- number of customers (n)
- facility fixed costs
- customer-to-facility assignment costs

Additionally, a companion file **uncapopt.txt** provides ground-truth optimal objective values, enabling exact benchmarking and validation.

3.2 Repository Organization

Raw data are stored in a dedicated repository-managed raw data directory:

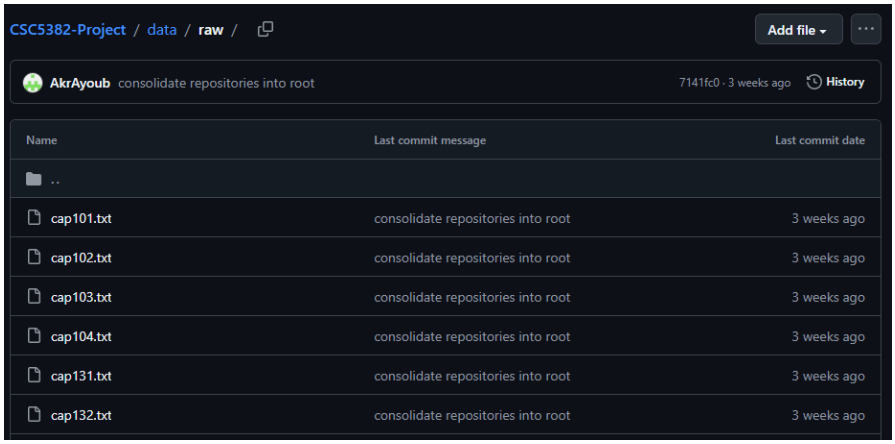


Figure 2: Raw data directory

3.3 Ingestion Pipeline Design

The ingestion stage is implemented in: **pipeline/ingest_data.py.** What the code does:

1. **Discovers dataset files**
   - scans **data/raw/** for all .txt instance files
   - filters out non-instance files (e.g., **uncapopt.txt**)
2. **Parses instance metadata**
   - number of facilities (m)
   - number of customers (n)
3. **Validates dataset completeness**
   - checks if each instance has a corresponding optimal value
4. **Builds dataset manifest**
   - creates structured metadata for all instances

3.4 Output Artifacts

The ingestion pipeline produces:

- dataset_manifest.csv
- dataset_manifest.json

These files contain:

- instance name
- dimensions (m, n)
- availability of optimal value
- parsing status

```
=== Milestone 3: Raw Data Ingestion Summary ===
Raw directory:[redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw
Optima file present: True
Total instance files discovered: 15
Successfully parsed headers: 15
Header parse errors: 0
Instances with known optimum entries: 15

Sample records:
- cap101: m=25, n=50, known_opt=True, status=ok
- cap102: m=25, n=50, known_opt=True, status=ok
- cap103: m=25, n=50, known_opt=True, status=ok
- cap104: m=25, n=50, known_opt=True, status=ok
- cap131: m=50, n=50, known_opt=True, status=ok

Manifest files written:
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\interim\dataset_manifest.csv
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\interim\dataset_manifest.json
PS[redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2>
```

Figure 3: Ingestion execution output

dataset_manifest.csv

Milestone_3-Data_Prep > data > interim > dataset_manifest.csv > data

```csv
instance_id,file_name,file_path,source,file_size_bytes,facility_count_m,customer_count_n,has_known_optimum,ingestion_status
cap101,cap101.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap101.txt,OR-Library UFLP,16057,25,50,True,ok
cap102,cap102.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap102.txt,OR-Library UFLP,16081,25,50,True,ok
cap103,cap103.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap103.txt,OR-Library UFLP,16081,25,50,True,ok
cap104,cap104.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap104.txt,OR-Library UFLP,16081,25,50,True,ok
cap131,cap131.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap131.txt,OR-Library UFLP,31815,50,50,True,ok
cap132,cap132.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap132.txt,OR-Library UFLP,31864,50,50,True,ok
cap133,cap133.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap133.txt,OR-Library UFLP,31864,50,50,True,ok
cap134,cap134.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap134.txt,OR-Library UFLP,31864,50,50,True,ok
cap71,cap71.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap71.txt,OR-Library UFLP,10445,16,50,True,ok
cap72,cap72.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap72.txt,OR-Library UFLP,10460,16,50,True,ok
cap73,cap73.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap73.txt,OR-Library UFLP,10460,16,50,True,ok
cap74,cap74.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\cap74.txt,OR-Library UFLP,10460,16,50,True,ok
capa,capa.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\capa.txt,OR-Library UFLP,1228450,100,1000,True,ok
capb,capb.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\capb.txt,OR-Library UFLP,1229598,100,1000,True,ok
capc,capc.txt,D:\aui\Ayoub\Spring_2026\AI for Digital Transformation\Project\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw\capc.txt,OR-Library UFLP,1228925,100,1000,True,ok
```

Figure 4: Ingestion manifest output

The ingestion stage transforms raw benchmark files into a structured and auditable dataset catalog, which is essential for reproducibility and pipeline automation.

4. Data Preprocessing & Transformation

4.1 Objective

The preprocessing stage transforms raw OR-Library text files into structured datasets suitable for analytics and feature engineering.

4.2 Parsing Challenges and Solutions

The OR-Library format contains inconsistencies such as: **capacity 2141200.** These values are irrelevant for UFLP but must be handled during parsing.

Solution implemented:

- ignore capacity tokens
- selectively extract fixed costs
- correctly align assignment cost matrix

4.3 Preprocessing Pipeline

Implemented in: **pipeline/preprocess_data.py**. What the code does:

1. **Reads raw tokens**
2. **Extracts structured components**
   - facility costs
   - customer costs
3. **Builds relational datasets**
4. **Writes outputs to CSV and JSON**

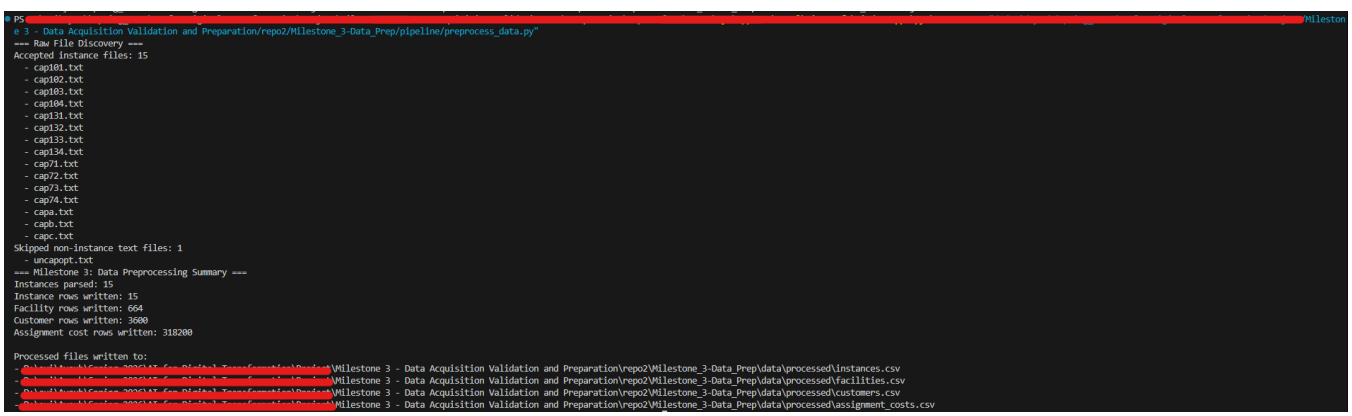


Figure 5: Preprocessing execution output

4.4 Output Tables and Scale

Table	Description
instances	instance-level metadata
facilities	facility costs
customers	customer identifiers
assignment_costs	cost matrix

Figure 6: Samples of processed datasets after preprocessing including normalized tables

The pipeline produces:

- 15 instances
- 664 facilities
- 3600 customers
- 318,200 assignment cost entries

5. Feature Engineering Design

5.1 Objective

Feature engineering transforms structured data into meaningful representations that can support:

- optimization analysis
- ML-based modeling
- feature reuse through a feature store

5.2 Feature Engineering Pipeline

Implemented in: **pipeline/engineer_features.py**. What the code does:

1. **Aggregates statistics**
   - mean, min, max, std
2. **Computes derived metrics**
   - cost ranges
   - ratios

3. **Normalizes features**
    - min-max scaling
    - z-score normalization
4. **Generates identifiers**
    - facility_key
    - customer_key
5. **Adds timestamps**
    - required for feature store integration

5.3 Feature Categories

- **Instance-Level**
    - facility_count_m
    - avg_fixed_cost
    - avg_assignment_cost
- **Facility-Level**
    - normalized cost
    - ranking
    - assignment statistics
- **Customer-Level**
    - nearest facility
    - assignment cost distribution

5.4 Feature Summary Table

Feature	Level	Description
avg_fixed_cost	Instance	Mean facility cost
fixed_cost_zscore	Facility	Normalized cost
nearest_facility_cost	Customer	Closest assignment cost

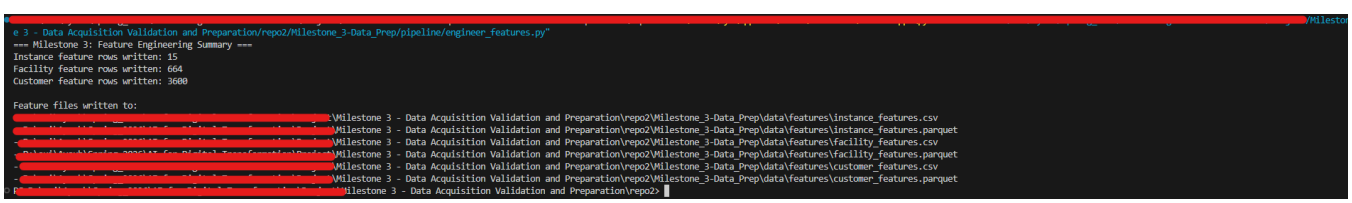


Figure 7: Features engineering execution output

customer_features.csv

Milestone_3-Data_Prep > data > features > customer_features.csv > data

customer_key,instance_id,customer_id,event_timestamp,min_assignment_cost,max_assignment_cost,avg_assignment_cost,std_assignment_cost,assignment_cost_range,nearest_facility_id,nearest_facility_cost
cap101__0,cap101,0,2026-03-21 00:00:00+00:00,146.0,58268.0,31574.48,25685.205486614275,58122.0,24,146.0
cap101__1,cap101,1,2026-03-21 00:00:00+00:00,87.0,11656.275,6657.795,2473.8948218345904,11569.275,24,87.0
cap101__2,cap101,2,2026-03-21 00:00:00+00:00,672.0,6919.7625,3475.7775,1507.092581587143,6247.7625,24,672.0
cap101__3,cap101,3,2026-03-21 00:00:00+00:00,1337.0,66973.2,24339.224,17014.615302598642,65636.2,24,1337.0
cap101__4,cap101,4,2026-03-21 00:00:00+00:00,31.0,142841.7375,50257.0645,40117.7042077478,142810.7375,24,31.0
cap101__5,cap101,5,2026-03-21 00:00:00+00:00,559.0,2512.9375,1499.7425,457.0060031607900 4,1953.9375,24,559.0
cap101__6,cap101,6,2026-03-21 00:00:00+00:00,2370.0,53852.6625,19529.8325,13613.169737978,51482.6625,24,2370.0
cap101__7,cap101,7,2026-03-21 00:00:00+00:00,1089.0,266358.375,103284.315,73829.43941612249,265269.375,24,1089.0
cap101__8,cap101,8,2026-03-21 00:00:00+00:00,33.0,108940.8375,38545.930499999995,31508.185059186224,108907.8375,24,33.0
cap101__9,cap101,9,2026-03-21 00:00:00+00:00,32.0,2869.7625,1697.1995000000002,555.4266482452927,2837.7625,24,32.0
cap101__10,cap101,10,2026-03-21 00:00:00+00:00,546.4,5495.0,1532.2,931.7845828301732,4948.6,6,546.4
cap101__11,cap101,11,2026-03-21 00:00:00+00:00,904.0,477721.5625,172202.7525,135705.09091160286,476817.5625,24,904.0
cap101__12,cap101,12,2026-03-21 00:00:00+00:00,1231.7,84557.9,28583.908,23954.52310100821,83326.2,3,1231.7
cap101__13,cap101,13,2026-03-21 00:00:00+00:00,143.0,139911.375,47988.633,40789.301058067125,139768.375,24,143.0
cap101__14,cap101,14,2026-03-21 00:00:00+00:00,615.0,15041.8125,5550.3345,3826.7466450976344,14426.8125,24,615.0
cap101__15,cap101,15,2026-03-21 00:00:00+00:00,564.0,54681.1875,23253.2625,12518.441537013503,54117.1875,24,564.0

facility_features.csv

Milestone_3-Data_Prep > data > features > facility_features.csv > data

facility_key,instance_id,facility_id,event_timestamp,fixed_cost,normalized_fixed_cost_minmax,fixed_cost_zscore,fixed_cost_rank_ascending,avg_assignment_cost_from_facility,min_assignment_cost_from_facility,max_assignment_cost_from_facility,std_assi
cap101__0,cap101,0,2026-03-21 00:00:00+00:00,58268.0,1.0,0.9592695390243703,13,47233.470499999996,1995.2,406770.0,71060.41938316799
cap101__1,cap101,1,2026-03-21 00:00:00+00:00,7500.0,0.1287155900322647,-1.0149047955635613,2,32685.8425,1326.2,325852.8,56253.56094100478
cap101__2,cap101,2,2026-03-21 00:00:00+00:00,58268.0,1.0,0.9592695390243703,14,29190.147999999997,693.5,322800.0,58170.99021868522
cap101__3,cap101,3,2026-03-21 00:00:00+00:00,7500.0,0.1287155900322647,-1.0149047955635613,3,26651.0925,924.825,264755.4,53604.4436525241
cap101__4,cap101,4,2026-03-21 00:00:00+00:00,58268.0,1.0,0.9592695390243703,15,29636.49,1063.525,294457.8,53925.1705330835
cap101__5,cap101,5,2026-03-21 00:00:00+00:00,7500.0,0.1287155900322647,-1.0149047955635613,4,51845.3085,1133.2,681269.4,104090.64181491744
cap101__6,cap101,6,2026-03-21 00:00:00+00:00,58268.0,1.0,0.9592695390243703,16,60595.2385,546.4,810550.8,120927.59457574626
cap101__7,cap101,7,2026-03-21 00:00:00+00:00,7500.0,0.1287155900322647,-1.0149047955635613,5,29515.234,1008.425,289434.6,55518.07263887752
cap101__8,cap101,8,2026-03-21 00:00:00+00:00,58268.0,1.0,0.9592695390243703,17,58766.581499999999,1406.8,728398.2,112487.54301805331
cap101__9,cap101,9,2026-03-21 00:00:00+00:00,7500.0,0.1287155900322647,-1.0149047955635613,6,25032.827,0.0,261790.8,51855.02720403149
cap101__10,cap101,10,2026-03-21 00:00:00+00:00,58268.0,1.0,0.9592695390243703,18,39693.261,768.8,512606.4,79553.96096795137
cap101__11,cap101,11,2026-03-21 00:00:00+00:00,7500.0,0.1287155900322647,-1.0149047955635613,7,98841.70199999999,1940.4,1361570.4,203553.93848562188
cap101__12,cap101,12,2026-03-21 00:00:00+00:00,58268.0,1.0,0.9592695390243703,19,38381.384,1145.2,451435.8,73499.60410452023
cap101__13,cap101,13,2026-03-21 00:00:00+00:00,7500.0,0.1287155900322647,-1.0149047955635613,8,65980.107,2314.8,644793.0,112076.62781749242
cap101__14,cap101,14,2026-03-21 00:00:00+00:00,58268.0,1.0,0.9592695390243703,20,50124.236500000006,1500.8,575875.2,92224.10087346504
cap101__15,cap101,15,2026-03-21 00:00:00+00:00,7500.0,0.1287155900322647,-1.0149047955635613,9,39360.85,1846.0,300835.0,62936.388290815616

Figure 8: Preview of engineered feature datasets including statistical, normalized, and derived attributes

6. Schema Definition

6.1 Objective

To ensure data consistency, integrity, and validation, explicit schemas were defined for each stage of the pipeline. These schemas act as contracts that enforce the expected structure and properties of the data at different stages of transformation.

6.2 Schema Layers

Three schema levels were implemented:

- raw_schema.json
- processed_schema.json
- feature_schema.json

Each schema corresponds to a specific pipeline stage:

Schema	Purpose
Raw	Defines structure of ingested metadata
Processed	Defines structured tabular outputs
Feature	Defines ML-ready feature datasets

6.3 Schema Structure

Each schema defines:

- required columns
- numeric columns
- non-negative constraints
- identifier columns

```
{} processed_schema.json ×
Milestone_3-Data_Prep > schema > {} processed_schema.json > {} facilities.csv > [] numeric_columns
1  {
2    "instances.csv": {
3      "required_columns": [
4        "instance_id",
5        "facility_count_m",
6        "customer_count_n",
7        "total_fixed_cost",
8        "avg_fixed_cost",
9        "min_fixed_cost",
10       "max_fixed_cost",
11       "total_assignment_cost_entries"
12     ],
13     "numeric_columns": [
14       "facility_count_m",
15       "customer_count_n",
16       "total_fixed_cost",
17       "avg_fixed_cost",
18       "min_fixed_cost",
19       "max_fixed_cost",
20       "total_assignment_cost_entries"
21     ],
22     "non_negative_columns": [
23       "facility_count_m",
24       "customer_count_n",
25       "total_fixed_cost",
26       "avg_fixed_cost",
27       "min_fixed_cost",
28       "max_fixed_cost",
29       "total_assignment_cost_entries"
30     ],
```

Figure 9: Example schema definition snippet for raw dataset specifying columns, fields, constraints...

6.4 What the Schema Enforces (Implementation-Level)

The validation system uses these schemas to:

- **Required Columns Check:** Ensures no missing essential data fields:
  - prevents incomplete datasets
  - avoids downstream failures (e.g., missing IDs)
- **Numeric Constraints:** Ensures columns expected to be numeric:
  - can be parsed as floats/ints
  - contain no invalid values
- **Non-Negativity Constraints:** Ensures logical correctness:
  - costs must be ≥ 0
  - counts must be ≥ 0
- **ID Integrity:** Ensures:
  - uniqueness of entities
  - consistency across joins

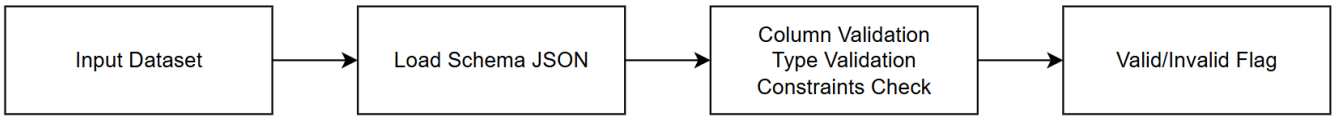


Figure 10: Schema validation pipeline

7. Data Validation & Anomaly Analysis

7.1 Objective

To verify that all datasets:

- conform to schema definitions
- contain no inconsistencies

- are statistically valid

7.2 Validation Pipeline

The validation is implemented in: **pipeline/validate_data.py**

7.3 Validation Logic

The validation system performs multiple checks:

1. **Schema Compliance**
   - verifies required columns exist
   - validates data types
2. **2. Numeric Checks**
   - ensures numeric columns contain valid values
   - detects invalid parsing
3. **3. Non-Negativity Checks**
   - ensures all cost values ≥ 0
   - ensures counts ≥ 0
4. **Row Consistency Checks**
   - verifies relationships across tables
   - ensures:
     - number of facilities matches metadata
     - assignment cost matrix is complete

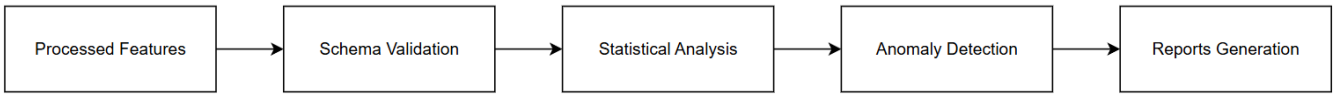
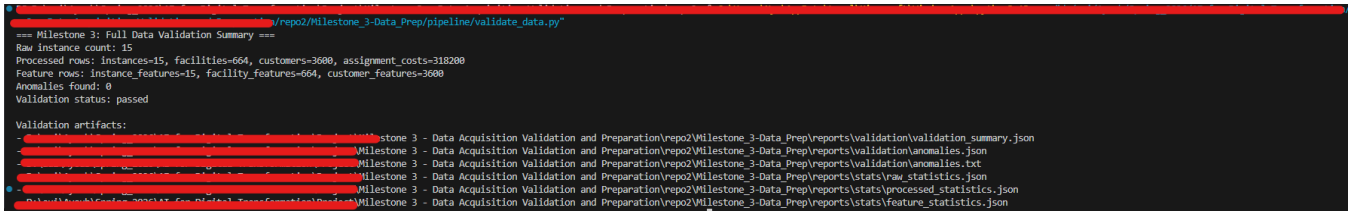


Figure 11: Data validation and anomaly detection pipeline

7.4 Outputs

Validation produces:

- validation_summary.json
- anomalies.json
- anomalies.txt
- statistical reports:
  - raw statistics
  - processed statistics
  - feature statistics



```
[redacted]/repo2/Milestone_3-Data_Prep/pipeline/validate_data.py"
=== Milestone 3: Full Data Validation Summary ===
Raw instance count: 15
Processed rows: instances=15, facilities=664, customers=3600, assignment_costs=318200
Feature rows: instance_features=15, facility_features=664, customer_features=3600
Anomalies found: 0
Validation status: passed

Validation artifacts:
- [redacted]stone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\validation\validation_summary.json
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\validation\anomalies.json
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\validation\anomalies.txt
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\stats\raw_statistics.json
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\stats\processed_statistics.json
[redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\stats\feature_statistics.json
```

Figure 12: Validation output summary showing zero anomalies across all datasets

8. Data Versioning with DVC

8.1 Objective

To ensure reproducibility and traceability of data transformations using Data Version Control (DVC).

8.2 DVC Setup

The repository was initialized with: **python -m dvc init**

8.3 Tracked Data Artifacts

The following directories are versioned:

- data/raw
- data/interim
- data/processed
- data/features
- reports/

8.4 DVC Execution and Flow:

DVC tracks:

- input dependencies (scripts + data)
- outputs (generated datasets)
- execution order (pipeline stages)

This enables the DVC to ensure:

- reproducible runs
- automatic dependency tracking
- versioned datasets

```yaml
stages:
  ingest: ...

  preprocess: ...

  features: ...

  validate:
    cmd: python pipeline/validate_data.py
    deps:
      - pipeline/validate_data.py
      - schema/raw_schema.json
      - schema/processed_schema.json
      - schema/feature_schema.json
      - data/interim/dataset_manifest.csv
      - data/interim/dataset_manifest.json
      - data/processed/instances.csv
      - data/processed/facilities.csv
      - data/processed/customers.csv
      - data/processed/assignment_costs.csv
      - data/features/instance_features.csv
      - data/features/facility_features.csv
      - data/features/customer_features.csv
    outs:
      - reports/validation/validation_summary.json
      - reports/validation/anomalies.json
      - reports/validation/anomalies.txt
      - reports/stats/raw_statistics.json
      - reports/stats/processed_statistics.json
      - reports/stats/feature_statistics.json
```

Figure 13: DVC pipeline definition showing ingestion, preprocessing, feature engineering, and validation stages

Note on the implementation:

Due to long file paths on Windows, DVC encountered path resolution issues, and a workaround was used by mapping the drive (subst X:) before executing the pipeline

9. Feature Store Setup: Feast

9.1 Objective

To enable centralized feature management and retrieval using a feature store.

9.2 Feature Store Components

- Entities
  - instance_id
  - facility_key
  - customer_key
- Feature Views
  - instance_features_view
  - facility_features_view
  - customer_features_view
- Data Source
  - Parquet files generated by pipeline

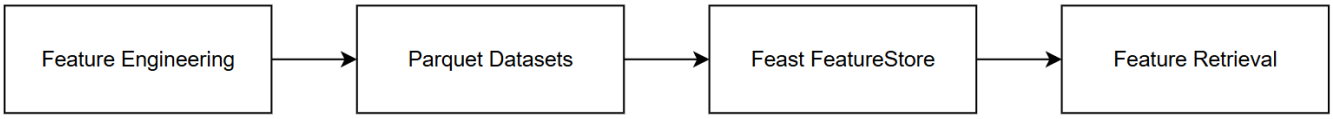
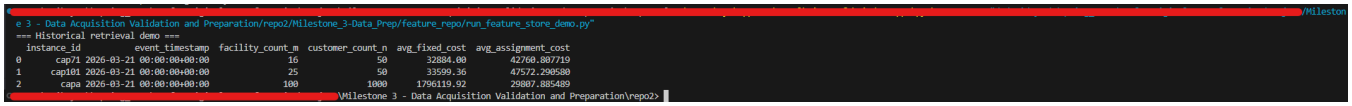


Figure 14: Feature store integration architecture

9.3 Retrieval Example

The system successfully retrieves:

- facility_count_m
- customer_count_n
- avg_fixed_cost
- avg_assignment_cost



```
[redacted]/Milestone
e 3 - Data Acquisition Validation and Preparation/repo2/Milestone_3-Data_Prep/feature_repo/run_feature_store_demo.py"
=== Historical retrieval demo ===
  instance_id          event_timestamp  facility_count_m  customer_count_n  avg_fixed_cost  avg_assignment_cost
0      cap71 2026-03-21 00:00:00+00:00                16                50        32884.00         42760.807719
1     cap101 2026-03-21 00:00:00+00:00                25                50        33599.36         47572.290580
2       capa 2026-03-21 00:00:00+00:00               100              1000      1796119.92         29807.885489
[redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2>
```

Figure 15: Feature Retrieval output of a sample showing instance-level features

10. Results & Demonstration: Pipeline Execution Results

- 15 instances processed
- 664 facilities
- 3600 customers
- 318,200 assignment cost entries
- 0 anomalies detected

```
run_full_workflow.py
Milestone_3-Data_Prep > run_full_workflow.py > main
1  from  future  import annotations
PROBLEMS   OUTPUT   DEBUG CONSOLE   TERMINAL   PORTS
[redacted]
idation and Preparation/repo2/Milestone_3-Data_Prep/run_full_workflow.py"rkflow.py";be78cd2d-08bc-41aa-869a-afff80bde580
Starting full Milestone 3 workflow...

================================================================================
RUNNING STEP: Milestone 3 Data Pipeline
================================================================================
Starting Milestone 3 full data pipeline...

================================================================================
RUNNING STEP: Raw Data Ingestion
================================================================================
=== Milestone 3: Raw Data Ingestion Summary ===
Raw directory: [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw
Optima file present: True
Total instance files discovered: 15
Successfully parsed headers: 15
Header parse errors: 0
Instances with known optimum entries: 15

Sample records:
- cap101: m=25, n=50, known_opt=True, status=ok
- cap102: m=25, n=50, known_opt=True, status=ok
- cap103: m=25, n=50, known_opt=True, status=ok
- cap104: m=25, n=50, known_opt=True, status=ok
- cap131: m=50, n=50, known_opt=True, status=ok

Manifest files written:
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\interim\dataset_manifest.csv
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\interim\dataset_manifest.json
COMPLETED: Raw Data Ingestion

================================================================================
RUNNING STEP: Data Preprocessing
================================================================================
=== Raw File Discovery ===
Accepted instance files: 15
  - cap101.txt
  - cap102.txt
  - cap103.txt
  - cap104.txt
  - cap131.txt
  - cap132.txt
  - cap133.txt
  - cap134.txt
  - cap71.txt
  - cap72.txt
  - cap73.txt
  - cap74.txt
  - capa.txt
  - capb.txt
  - capc.txt
Skipped non-instance text files: 1
  - umcapopt.txt
=== Milestone 3: Data Preprocessing Summary ===
Instances parsed: 15
Instance rows written: 15
Facility rows written: 664
Customer rows written: 3600
Assignment cost rows written: 318200

Processed files written to:
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\processed\instances.csv
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\processed\facilities.csv
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\processed\customers.csv
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\processed\assignment_costs.csv
COMPLETED: Data Preprocessing

================================================================================
RUNNING STEP: Feature Engineering
================================================================================
=== Milestone 3: Feature Engineering Summary ===
Instance feature rows written: 15
Facility feature rows written: 664
Customer feature rows written: 3600
Feature files written to:
[redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\features\instance_features.csv
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\features\instance_features.parquet
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\features\facility_features.csv
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\features\facility_features.parquet
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\features\customer_features.csv
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\features\customer_features.parquet
COMPLETED: Feature Engineering

================================================================================
RUNNING STEP: Data Validation
================================================================================
=== Milestone 3: Full Data Validation Summary ===
Raw instance count: 15
Processed rows: instances=15, facilities=664, customers=3600, assignment_costs=318200
Feature rows: instance_features=15, facility_features=664, customer_features=3600
Anomalies found: 0
Validation status: passed

Validation artifacts:
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\validation\validation_summary.json
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\validation\anomalies.json
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\validation\anomalies.txt
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\stats\raw_statistics.json
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\stats\processed_statistics.json
- [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\stats\feature_statistics.json
COMPLETED: Data Validation

================================================================================
MILESTONE 3 DATA PIPELINE COMPLETED SUCCESSFULLY
================================================================================

Generated artifact locations:
- Raw data:        [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\raw
- Interim data:    [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\interim
- Processed data:  [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\processed
- Features:        [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\data\features
- Validation:      [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\validation
- Statistics:      [redacted]\Milestone 3 - Data Acquisition Validation and Preparation\repo2\Milestone_3-Data_Prep\reports\stats
COMPLETED: Milestone 3 Data Pipeline

================================================================================
RUNNING STEP: Feast Repository Apply
================================================================================
Feast repository applied successfully.
COMPLETED: Feast Repository Apply

================================================================================
RUNNING STEP: Feast Feature Retrieval Demo
================================================================================
=== Historical retrieval demo ===
  instance_id           event_timestamp  facility_count_m  customer_count_n  avg_fixed_cost  avg_assignment_cost
0       cap71 2026-03-21 00:00:00+00:00                16                50        32884.00         42760.807719
1      cap101 2026-03-21 00:00:00+00:00                25                50        33599.36         47572.290580
2        capa 2026-03-21 00:00:00+00:00               100              1000      1796119.92         29807.885489
COMPLETED: Feast Feature Retrieval Demo

================================================================================
FULL MILESTONE 3 WORKFLOW COMPLETED SUCCESSFULLY
================================================================================
```

Figure 16: Full pipeline run output

12. Limitations & Next Steps

12.1 Limitations

- Synthetic timestamps (not real-time)
- Static dataset (no streaming ingestion)
- Windows-specific DVC issues

12.2 Future Work

- real-time data ingestion
- dynamic feature updates
- integration with model training
- cloud deployment (MLOps platform)
- direct integration of feature retrieval into solver-facing experimentation
- automated retriggering of validation after raw data changes